LINEAR ALGEBRA · LESSON 3

Multiplication and Inverse Matrices

Five ways to see a matrix product. Then the inverse: when it exists, and how **Gauss–Jordan** finds it.

Shown on real money questions: many portfolios, hedges and option payoffs.

AB five ways | Gauss–Jordan | MIT 18.06, Lecture 3

THE SETUP

Many clients, one set of funds

Stack each client's portfolio as a column of X. One product gives every client's budget and return:

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_{A\text{: funds}} \underbrace{\begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix}}_{X\text{: clients}} = \begin{bmatrix} 10 & 10 \\ 105 & 90 \end{bmatrix}$$

Then the reverse question: given the targets, which portfolios? That's the **inverse.**

WAY 1

Row times column

Entry (i, j) = row i of A · column j of B:

$$c_{22} = 12 \times 4 + 7 \times 6 = 90$$

Client 2's return: ₹4 lakh Nifty earns 48, ₹6 lakh bonds earns 42.

> **RULE · SHAPES**
>
> $(m \times n)(n \times p) = m \times p$. Inner sizes must match.

WAYS 2 AND 3

By columns, by rows

Columns of AB combine columns of A: one column per client.

$$A\begin{bmatrix}7\\3\end{bmatrix} = 7\begin{bmatrix}1\\12\end{bmatrix} + 3\begin{bmatrix}1\\7\end{bmatrix} = \begin{bmatrix}10\\105\end{bmatrix}$$

Rows of AB combine rows of B: one row per rule.

$$12\,(7,4) + 7\,(3,6) = (105, 90)$$

WAY 4

Columns times rows

A column times a row is a whole matrix. AB is a sum of them, one per fund:

$$\underbrace{\begin{bmatrix} 7 & 4 \\ 84 & 48 \end{bmatrix}}_{\text{Nifty's part}} + \underbrace{\begin{bmatrix} 3 & 6 \\ 21 & 42 \end{bmatrix}}_{\text{bonds' part}} = \begin{bmatrix} 10 & 10 \\ 105 & 90 \end{bmatrix}$$

IN MARKETS

Attribution by source: what each fund contributes to every client.

WAY 5

Block multiplication

Cut the matrices into blocks that match, then multiply blocks like numbers:

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} \rightarrow \text{top-left} = A_1B_1 + A_2B_3$$

Three funds: "Nifty + bonds" part plus "midcap" part. Big computers multiply this way.

INVERSES

When can A be undone?

$A^{-1}A = I = AA^{-1}$. But two identical Nifty funds:

$$\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

₹1 lakh more in one, ₹1 lakh less in the other: **a trade that changes nothing.**

RULE · SINGULAR TEST

$Ax = 0$ with $x \neq 0 \Rightarrow$ no inverse.

GAUSS–JORDAN · 2 × 2

[A | I] → [I | A⁻¹]

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 12 & 7 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & -5 & -12 & 1 \end{array}\right]$$

Gauss goes down. **Jordan** goes up, then divide by the pivots:

$$\left[\begin{array}{cc|cc} 1 & 0 & -1.4 & 0.2 \\ 0 & 1 & 2.4 & -0.2 \end{array}\right]$$

GAUSS–JORDAN · 3 × 3

The three-fund portfolio

After the downward half, the right side is Lesson 2's elimination matrix E, messy 1.4 and all. Then up, then divide:

$$A^{-1} = \begin{bmatrix} 4.2 & -0.6 & 4 \\ 0.3 & 0.1 & -1.5 \\ -3.5 & 0.5 & -2.5 \end{bmatrix}$$

WHY IT WORKS

$E[A \mid I] = [EA \mid E]$. When $EA = I$, E must be A^{-1}.

READING A^{-1}

Columns = building-block trades

Column	Trade (Nifty, bonds, midcap)	Effect
1	$(4.2, 0.3, -3.5)$	+₹1 lakh budget only
2	$(-0.6, 0.1, 0.5)$	+₹1k return only
3	$(4, -1.5, -2.5)$	+1 exposure only

Want ₹5k more return, nothing else changed? Trade $5\times$ column 2.

Arrow–Debreu building blocks

For futures + 19,900 call + 20,100 call, the columns of A^{-1} give lot mixes that pay in **exactly one** expiry scenario:

- Sell 1 futures, buy 2 × 20,100 C: pays 200 only at 19,800
- Buy 1 × 19,900 C, sell 3 × 20,100 C: pays 100 only at 20,000
- Buy 1 × 20,100 C: pays 100 only at 20,200

Any payoff over these three levels is a mix of these.

RECAP

Key takeaways

1. AB by entries, columns, rows, column×row, or blocks: same answer.
2. No inverse if some trade changes nothing ($Ax = 0$).
3. Gauss–Jordan: $[A \mid I] \to [I \mid A^{-1}]$.
4. $X = A^{-1}B$ solves every target at once.
5. Columns of A^{-1} are building blocks.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A lab that runs Gauss–Jordan on your own matrix, step by step, plus worked hedging and options examples:

learn.krisarj.com

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